Project Proposal

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Abstract

Maternal malnutrition in rural Limpopo is compounded by poor connectivity, high data costs, and a lack of culturally relevant guidance, leaving pregnant women and their support structures without effective nutritional support. AMEN is a lightweight, offline-first Android application developed to address this gap, delivering evidence-based maternal nutritional guidance through multilingual audio-visual content, digital storytelling, and role-based design tailored to pregnant women, community health workers, and family members. Its specifications and content draw from collaborative research by the University of Limpopo (UL), University College Dublin (UCD), and the AMEN project team. The project progresses through development, UAT, and iterative redesign, with the resulting application representing a tangible contribution to accessible maternal nutrition support in rural Limpopo. Usability is measured against the ISO 9241-11 framework alongside a pre- and post-interaction knowledge assessment, with findings expected to contribute evidence on the relationship between usability and knowledge gain in low-resource mHealth contexts.

Keywords

Maternal Health, mHealth, User-Centred Design, Low-Resource Settings, GRAVID-Adapt, Community Healthcare Workers (CHWs)

1 Introduction

Despite recent improvements in child and infant mortality across Sub-Saharan Africa, the area still lags behind developed countries, highlighting the continuing need to find solutions to address this issue [1]. Malnutrition in pregnant women in lower-middle-income countries (LMICs) remains a cause of concern, as tension often arises between accepted cultural practices and evidence-based Western dietary recommendations [5, 15]. Mobile health interventions currently available in this context, such as MomConnect and Pregnancy+, have shown promise but fall short in cultural adaptation, effective nutritional focus, and accessibility, as many devices lack the storage capabilities for data-dependent applications [1, 13, 14]. Mothers also struggle to navigate complex or text-heavy applications, particularly those not available in their home language [13].

Collaborative research by the AMEN project team, University College Dublin (UCD), and the University of Limpopo (UL) identified the need for a lightweight, offline-first, culturally sensitive nutrition application with multilingual and audio-visual support. To address this gap we will develop a mobile application that uses digital storytelling, trusted community voices, and role-based design to deliver maternal nutritional guidance to pregnant women and their support structures in rural Limpopo. Its design is informed by focus groups and co-design sessions with the target community, ensuring every decision is grounded in the realities of its users. The project will progress through design, development, user acceptance testing (UAT), and co-design, feedback evaluation, and redesign phases. Ultimately, AMEN aims to demonstrate that technology designed with a community, not just for one, can make a real difference to maternal health outcomes.

2 Related Work

2.1 Maternal mHealth in South Africa and the Limpopo context

Scoping reviews consistently find that mHealth interventions in LMICs lack community involvement and cultural adaptation. [1, 17]. An interesting example is South Africa's MomConnect, which demonstrated real demand and enrolled over 494,000 women in its first year [3]. However, its USSD and WhatsApp architecture meant that it has limited functionality and scaled poorly in terms of cost, leading its own authors to suggest the development of a custom mobile application to enable multimedia delivery [14]. Another example is Pregnancy+, which is used in South Africa but presents storage and connectivity barriers that exclude many rural users [12]. Muthelo et al.'s research in rural Limpopo most directly motivates this project [12]. Workshops conducted identified storage limits, data costs, and poor connectivity as the primary barriers to existing applications [12]. Participants stated preference for video and audio over text and recommended that fathers be trained alongside mothers [12].

2.2 Co-design and community involvement

Coleman et al. presented findings from generative co-design workshops across South African sites including rural Limpopo and reported that communities welcome digital interventions, provided that they are built in local languages and cultural context [6]. A key finding was that automated translation cannot be relied on for South African languages, and first-language speakers must be involved in content creation from the start. Till and Farao's reflections on workshops across four South African provinces surface a tension our UAT must navigate [16]. In Limpopo, the presence of male community leaders altered group dynamics, with female and younger participants often deferring rather than contributing freely. The authors suggest separating men and women for sensitive health topics, which conflicts with Muthelo's call to include fathers [12].

2.3 Trusted voices and information sources

Pregnant women in low-resource settings navigate conflicting advice from healthcare professionals, family elders, traditional healers,

and digital tools [7, 12]. Dsane-Nsor et al. found that women in Ghana rely heavily on older female relatives for antenatal information and that testimonials and narrative storytelling strengthen trust in digital content [7]. Félix et al. found that parents actively filter health messages based on whether the source feels competent, values-aligned, and connected to their lived reality, dismissing generic content even from healthcare professionals [9]. Farao's digital storytelling argument sits alongside these findings, proposing that stories grounded in existing cultural practices help mothers move from passively receiving information to acting on it [8].

2.4 Evaluating mHealth in low-literacy contexts

Molefe and Dlamini provide quantitative evidence that user-centred redesign measurably improves mHealth outcomes, with adoption rising from 41.2% to 72.4% and 90-day retention from 29.7% to 51.8% in a South African context [10]. Till et al. argue that UAT in this context needs working MVPs rather than wireframes and also importantly found that design cards support participants with varying literacy levels [18]. Atukunda et al.'s mixed-methods approach which combined structured tasks on a working pilot with qualitative follow-up provides a good working example of what we believe will be an effective evaluation pattern [2]. The predefined tasks further lend themselves to structured quantitative measurement through the ISO 9241-11 usability framework, which defines usability as "the extent to which a system, product or service can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use" [4]. Evaluating usability across these dimensions, always relative to the specific context of use, makes it particularly suited to mHealth applications designed for constrained deployment environments.

3 Problem Statement

A critical gap in maternal nutritional knowledge in rural Limpopo drives widespread under-nutrition and directly contributes to a maternal mortality rate that exceeds the global average at 536 deaths per 100,000 live births [11]. There is currently no effective means by which this knowledge is provided to the mothers and their support systems in the rural Limpopo context.

Current maternal health applications are either too simplistic and leave the users feeling disconnected and overwhelmed by text, or they are too complex and demand technological infrastructure and literacy levels that exceed what is realistically available in this context. They do not promote a maternal diet that is financially and culturally sustainable for individuals in the context of rural Limpopo.

This project aims to bridge that gap through the development of an application grounded in field evidence and the lived realities of its target users.

4 Aims and Research Question

(1) Develop an MVP for the UAT, which is built according to the Product Requirements Document provided by the proposer
(2) Produce an iterated design grounded in user testing findings and analysis of the qualitative data, reflecting genuine community collaboration
(3) Deliver a culturally-adapted, evidence-based, and offline-first maternal nutrition application that is stable for pilot deployment and can be iterated on by future project contributors

4.1 Research Question

Beyond these aims, the project's methodology provides an opportunity to explore the following research question:

(1) Does the nutritional knowledge retained from interacting with a maternal mobile health application correlate with the perceived usability of that application among pregnant women and their support structures in rural Limpopo?

5 Procedures and Methods

5.1 Design Features

The principal features of the application are summarised in Table 1.

Table 1: Principal design features of the AMEN application.

Feature	Description
Role-based experiences	Three distinct experiences (pregnant woman, CHW, and family member) within a single cohesive application. Role is selected at onboarding and drives the home screen, visible content, and supporting features.
Modular content viewer	Text, audio, video, and interactive myth-busting cards will be displayed. New or revised content requires no code change so easy to add for administrators.
Trimester-aware content and nudges	Users select their weeks since conception at onboarding. The home screen surfaces trimester-relevant modules, frequent learning nudges are drawn from trimester-matched content, and the final build's download-on-demand bundles are organised by trimester.
Recipes and meal preparation	A dedicated recipe stream combines short video demonstrations with illustrated step-by-step cards, using locally available ingredients identified in the AMEN research team's thematic work. Recipes are trimester-tagged and surfaced alongside the corresponding nutritional content.
Reminders and appointments	User-configurable reminders for antenatal care visits, supplement intake, and daily learning nudges. Reminders are local to the device so they function without connectivity, and their schedule adapts to the user's current trimester.
Bilingual support (English and Sepedi)	All user-facing strings are available in both languages. Language is selected at onboarding and can be changed at any time from the settings screen.
Offline-first user data	User role, trimester, language, progress, and reminder schedules are stored locally on the device and synchronised to the backend when connectivity is available.

5.2 Development Platform

The application will be developed in React Native, targeting Android. React Native was selected over native Android development for its cross-platform extensibility to iOS in future iterations as well as for its strong ecosystem of libraries suited for rapid development. The user interface is built using the Gluestack UI component library, which provides accessible primitives out of the box and makes development easier.

The backend uses Supabase, which provides authentication, a PostgreSQL database with row-level security, and object storage for

media content in a single managed service. It was also selected for its speed of development and support for the offline-sync patterns. Local on-device storage uses Expo sqlite for structured data. Crash reporting will use Sentry during development and the pilot, with lightweight anonymous event analytics added in the final phase. The codebase is managed in a private GitHub respository.

5.3 Implementation Strategy

We will carry out application development backend first, with the API, data layer, and content access infrastructure built before UI implementation begins. This allows each screen to be wired against working endpoints as the UI prototypes are delivered. Development will be iterative with weekly supervisor check-ins and goals. Testing uses unit tests on the data layer and integration tests on the offline-first synchronisation. We will utilize an Android emulator and a physical low-specification device from early in development. The development will happen in two stages. The MVP build will embed all educational content in the APK so the application works without network connectivity at the focus group venue. Then the final build replaces this with a download-on-demand architecture: the APK stays under 50 MB, and on first launch the user selects their current trimester and the application downloads the corresponding content bundle from Supabase Storage.

5.4 Expected Challenges

Implementing the two different content delivery strategies is the largest architectural challenge. The MVP build must work reliably without any network at the venue and the final build must handle partial downloads and interruptions.

Designing for variable literacy users in a culturally specific context also presents a significant challenge and cuts across every design decision. The application cannot assume reading fluency, familiarity with mobile interaction patterns, or cultural neutrality in its visual design.

Designing that application for a variety of devices also presents a challenge. Focus group participants will bring a wide variety of Android devices of varying age and specification, alongside a small number of team-supplied devices.

Offline-first synchronisation of user data is a smaller but real challenge. User state is held on-device and synchronised to Supabase when connectivity is available which requires a local-first write pattern with a background push queue rather than assuming every write reaches the backend immediately.

5.5 User Interaction Plan

Participants will be recruited with the assistance of UL researchers across three clinics in the Limpopo area, with focus groups held within each clinic per user group separately.

Table 2: Focus group participant breakdown.

User Group	Per Group	Clinics	Total
Community health workers	6	3	18
Pregnant mothers	8	3	24
Family decision-makers	6	3	18
Total			**60**

The group sizes are deliberately kept small, as Till et al. identify intimate group settings as more effective for community co-design and participant engagement [18].

A language barrier exists for the UCT research team, which directly shapes the facilitation structure of the sessions. Ahead of the trip, online and in-person familiarisation sessions will be held with the UL researchers and translator, as outlined in Table 3.

At each clinic, the CHW focus group will be conducted first. This serves a dual purpose: it gives the CHWs exposure to the application ahead of their role as trusted intermediaries in later focus groups and digital champions after deployment of the application, and it builds familiarity and trust between the CHWs and the research team. The UL researchers will lead facilitation throughout, with the UCT team and translator present across all sessions for support. Sessions will be voice-recorded and observational notes taken throughout.

Table 3: User-Focused Work Timeline.

Timeline	Description
20 April–6 July	Iterative UI and workshop design through weekly check-ins with the UL team and project proposer.
13 July	Online familiarisation session with UL researchers and translator covering the workshop structure, process, and application.
20 July	In-person familiarisation session between the UCT and UL teams covering the workshop structure, process, and application.
21 July	Clinic 1. CHW focus group (2 hrs), followed by pregnant mothers focus group (2 hrs), followed by family decision-makers focus group (2 hrs).
22 July	Clinic 2. CHW focus group (2 hrs), followed by pregnant mothers focus group (2 hrs), followed by family decision-makers focus group (2 hrs).
23 July	Clinic 3. CHW focus group (2 hrs), followed by pregnant mothers focus group (2 hrs), followed by family decision-makers focus group (2 hrs).
16 August	Evaluation of redesigned application by UL researchers. Repeat part 1 of focus group session structure.

5.6 User Focus Groups: Session Structure

The Focus groups will start with a discussion about the purpose of the sessions and the importance of maternal nutrition. Then, focus groups will follow a two-part layout:

Part 1 - Prototype interaction: Participants will complete a ten-question maternal nutrition knowledge assessment. Each question is intentionally aligned with content embedded within the application's features (e.g., "What makes a nutritious breakfast for a pregnant mother?"). Participants will then be presented with the high-fidelity prototype of the application and will have 10 minutes

to familiarise themselves with it. Participants will then be asked to complete a set of basic, predefined tasks to navigate the application's features, during which they will encounter content that addresses the knowledge assessment questions. During these tasks, the UCT and UL researchers will record the following measures in accordance with the ISO 9241-11 usability framework:

- **Effectiveness** — task completion rate; whether the participant successfully completed each task without assistance
- **Efficiency** — task completion time; the time taken to complete each task
- **Satisfaction** — recorded through the Likert-scale questionnaire completed following the interaction

Participants will then complete the knowledge assessment again. The results will be translated and evaluated against the application's nutritional content as follows:

Table 4: Knowledge assessment scoring criteria.

Response	Score	Criteria
Incorrect	0	Not inline with application content
Partly correct	0.5	Aspects of application content are present
Correct	1	Mostly/Completely inline with application content

Part 1 serves as a technology probe, familiarising participants with the application before Part 2 begins, limiting the blank page phenomenon and enabling more meaningful engagement with the co-design activity.

Part 2 - Co-design: Two different UI designs with minimal interactive ability will be presented after the prototype to act as further inspiration, giving participants a varied basis for comparison rather than limiting them to a single visual reference. Participants will then engage in a prompted co-design activity using design cards and office supplies, allowing them to reflect on all three designs and construct their own vision of an ideal interface and content delivery method. This shifts the dynamic from passive evaluation to active participation, giving the community genuine agency in shaping the final design.

5.7 User Data Evaluation and Redesign

The change in scores between the pre- and post-interaction knowledge assessments will provide a measure of content delivery effectiveness, contributing to the final success measures and the research question. Additional quantitative measures will be gathered in line with the ISO 9241-11 framework. These measures establish the magnitude of usability issues and inform the scale and priority of redesign decisions. These are not the primary driver of redesign but provide quantitative grounding for the qualitative insights that follow.

The qualitative insights gathered through voice recordings, researcher notes, co-design card outputs, and semi-structured discussions will be analysed using Thematic Analysis through inductive pattern recognition and deductive searches for feasible ideas. This data will also act to contextualise the quantitative results, for example, if task completion rates are low, the qualitative findings will help identify why and point toward design adjustments.

5.8 Final Product Success Measures

Functional and Non-Functional Requirements Evaluation: An evaluation against the functional and non-functional requirements outlined in the project specifications document will be conducted. Each requirement will be assessed as met, partially met, or not met, providing an account of the system's capabilities at the point of delivery. Functional requirements will be verified through direct feature testing, and non-functional requirements through performance testing on low-specification Android devices.

Usability and Content Effectiveness: Success will be defined as measurable improvement across all three ISO 9241-11 dimensions: improved task completion rates, reduced completion times and error rates, and improved satisfaction scores. Improved pre- to post-interaction knowledge assessment scores will additionally indicate more effective content delivery. Together, these results will provide a quantitative context for the research question.

These assessments determine whether the system does what it is supposed to do and in a way that is genuinely accessible, usable, and informative for the people it was designed for.

6 Ethical, Professional, Legal Issues

6.1 Ethical Considerations

Ethics clearance will be obtained before conducting the field work in July. We acknowledge that pregnant women are recognized as a vulnerable population and all research will be conducted in such a way that prioritizes their welfare and needs.

Written, informed consent will be obtained from every participant before the research sessions begin. The consent forms will be provided in English and Sepedi and be read aloud in order to accommodate varying literacy levels. Participants will be informed at the start of the session that they are able to withdraw at any time with no consequence and do not have to provide reasoning.

We also acknowledge the power dynamic and potential language barrier between us as university researchers and the participants, and will receive assistance from our collaborators at the University of Limpopo and their CHW connections in facilitating successful sessions.

6.2 Professional Considerations

Care will be taken to align with the Product Requirements Document provided by our proposer, and key decisions that require additional guidance not provided in the document will be made in consultation with her.

Findings will be available to the project's research partners and all contributors will be rightfully acknowledged in the final paper and project artefacts.

6.3 Legal Considerations

The application collects personal information and therefore falls under the Protection of Personal Information Act (POPIA). We have ensured that the app will only collect what it needs to function, and

no medical or diagnostic information will be collected. Backend access is restricted so users can only see their own data and local storage is encrypted on the device. All traffic between the app and the backend will be over HTTPS.

We also aim to make it clear that the app is an educational tool, not a medical device. The application only delivers content screened by the AMEN research team and a disclaimer on first launch and in the settings screen will tell users the app is educational and not a replacement for seeing a healthcare provider.

Intellectual property in the code and research outputs belongs to the University of Cape Town, in line with UCT's standard policy for supervised student research. The codebase will be available after submission and may be easily made available to future students continuing the AMEN research. Educational content in the app such as modules, videos, audio, and translations is owned by the AMEN research group.

7 Anticipated Outcomes

7.1 System

Two builds of the AMEN application will be produced: a pilot build embedding all content directly in the APK for the Limpopo field study, and a final build of under 50 megabytes using download-on-demand content for submission.

We aim to overcome the earlier identified design challenges and utilize our findings from the UAT workshop to successfully deliver the following:

A two-phase content delivery architecture. The application will support embedded and download-on-demand content.

An interface revised against real user findings. The final build will incorporate the interface changes identified as necessary in the Limpopo pilot.

A field-deployable Android application. By completion, the application will have been installed and used on the range of heterogeneous Android devices that participants brought to the Limpopo workshops, demonstrating cross-device compatibility under realistic field conditions rather than under lab conditions alone.

An offline-first user experience. The completed system will allow users to install, open, and engage with every feature of the application without a network connection, synchronising to the backend when connectivity is available. This is a concrete differentiator from the existing maternal health applications reviewed in the related work.

7.2 Expected Impact

This project's impact is most direct on the AMEN research programme, which gains a working application that future contributors can extend and improve. For the focus group participants, the application offers nutritional knowledge that they can begin to apply in their own pregnancies and hopefully share with the community.

7.3 Key Success Factors

Project success will be judged against four criteria:

Requirements met. The application is assessed against the functional and non-functional requirements in the project specifications, with each marked as met, partially met, or not met.

Table 5: Principal risks and mitigation strategies.

Risk	Impact	Mitigation
Late content delivery from AMEN/UL	Critical	Schedule locked early, placeholders unblock development, weekly check-in.
UI deliverables take longer than expected	High	Weekly design progress check-in, Gluestack defaults as placeholders if needed.
Exam period compresses capacity more than planned	High	Feature work front-loaded before exams, post-exam weeks treated as intense work sprints.
App fails on participants' devices in Limpopo	High	Cross-device testing from early development; team-supplied fallback devices.
Focus group disruption on the day	High	Printed screenshot backups; ability to skip broken features
Some features not developed by feature freeze date	High	Documented cut order of features
Sepedi reviewer turnaround longer than planned	Medium	Reviewer lined up early; English-only fallback for workshop if corrections late.

Successful field deployment. The application installs and runs on the range of Android devices participants bring to Limpopo and supports the planned activities without network connectivity at the venue.

Measurable knowledge gain. A meaningful improvement between pre- and post-interaction assessment scores indicates the content and its presentation are working as intended.

Usability against ISO 9241-11. The structured UAT tasks and qualitative feedback are assessed across effectiveness, efficiency, and satisfaction.

8 Project Plan

8.1 Risks and Risk Management Strategies

The greatest risk is the content delivery from the AMEN/UL research team. We will mitigate this by establishing contact early and use placeholders to unblock development.

The most consequential risks and their mitigations are summarised in Table 5.

8.2 Timeline

The project runs from 20 April 2026 to 26 October 2026, structured as six work phases followed by the post-submission deliverables period. The phases are summarised below, and the full schedule is given as a Gantt chart in Figure 1.

Foundations (20 April – 3 May)

Initial technical foundations like the React Native project, Supabase backend, authentication, sync prototype are laid before the proposal is submitted.

Core build (4 May – 24 May)

The application shell, content viewer, and settings screen are built. Sarah produces the navigation and home page Figma by 17 May, which unblocks the UI implementation. Workshop methodology design run in parallel.

Exam period (25 May – 7 June)

Both team members are at reduced capacity. Light work only: bug fixes, administrative tasks, and workshop logistics.

Intense push to freeze (8 June – 21 June)

Sarah's full core screens Figma is delivered on around 8 June, and the remaining application features are built against it. Sepedi strings are sent to the reviewer, integration and cross-device testing complete, and the feature freeze is reached on 21 June.

Polish and workshop readiness (22 June – 19 July)

Holiday-pace polish, content quality assurance as materials arrive from AMEN/UL and the Sepedi reviewer, and dress rehearsals of both the application and the focus group facilitation. Focus group materials such as the session plan, consent forms in both languages, and co-design materials are finalised by 5 July.

Limpopo field study and synthesis (Approx. 20 July)

The user acceptance testing study in Limpopo takes place in the week of 20 July. The Project Progress Demonstration follows on 27 July, and thematic analysis of the qualitative data alongside SUS and pre/post knowledge scoring begins immediately.

Post-pilot iteration and paper (3 August – 14 September)

Two parallel tracks run: a user interface iteration informed by pilot findings, and the Phase 2 download-on-demand architecture needed for the final build. Paper drafting is integrated with both tracks. The paper scaffold is due 10 August, the full draft 24 August, the final paper 4 September, and the final code submission 14 September.

Post-submission deliverables (15 September – 26 October)

The final demonstration window (16–23 September), poster (28 September), project website (5 October), and the School of Information Technology Showcase (26 October, to be confirmed).

8.3 Resources Required

People: the two-developer team, the project supervisor, the AMEN/UL research team as project proposers, community health workers helping facilitate the Limpopo focus groups, a Sepedi language reviewer.

Software: React Native, Gluestack UI, Supabase, op-sqlite, Figma, GitHub, Sentry.

Hardware: two to three Android test devices spanning a range of specifications, with at least one low-RAM device below two gigabytes to match the non-functional requirements; audio recording equipment for workshop data capture; team-supplied pre-loaded devices for any workshop participant whose phone cannot install the application.

Content: Nutrition modules with audio and video, the Sepedi explainer video for onboarding, myth-busting materials, trusted voices recordings, and Sepedi translations of application strings.

8.4 Deliverables

The project produces fifteen dated deliverables across the project lifetime, listed in Table 6.

Table 6: Project deliverables.

Date	Deliverable	Owner
17 May	Navigation and home page Figma	Sarah
8 June	Full core screens Figma (interactive)	Sarah
14 June	Application feature-complete (English); Sepedi strings out for review	Ella
5 July	Workshop materials complete (consents, SUS, pre/post knowledge tool, co-design materials, facilitation guide)	Sarah
19 July	Pilot build (pre-baked APK) ready for installation	Ella
20–24 July	Limpopo user acceptance testing	Both
27 July	Project Progress Demonstration	Both
10 August	Paper scaffold	Both
24 August	Full paper draft	Both
4 September	Final paper submitted	Both
14 September	Final code submission (APK under 50 MB, download-on-demand)	Ella
16–23 September	Final demonstration	Both
28 September	Academic poster	Both
5 October	Project website	Both
26 October (TBC)	School of IT Showcase	Both

8.5 Work Allocation

Ella owns the application architecture, backend, and UI integration: technical foundations, authentication, synchronisation, APK bundling, the two-phase content delivery strategy, cross-device testing, and full implementation of the UI against Sarah's prototypes. She leads the systems sections of the paper (system architecture, implementation, technical methodology). Her standalone contribution is an offline-first mHealth architecture with two-phase content delivery, validated in field conditions.

Sarah owns the user interface design and field research: interactive Figma prototypes for the full UI, focus group methodology, lead facilitation in Limpopo, and post-pilot analysis. She leads the user-research sections of the paper (UAT methodology, ISO 9241-11 findings, pre/post knowledge findings, co-design analysis). Her standalone contribution is a user-centred interface design and mixed-methods evaluation methodology combining ISO 9241-11 usability with pre/post knowledge gain.

9 Conclusion

The literature on maternal mHealth in low-resource settings points to a consistent set of requirements: cultural relevance, community involvement, multimedia content in local languages, and design that recognises the broader support network around pregnant women. Our application will be designed to directly address these requirements. Its offline-first, role-based architecture allows the application to function in conditions of variable connectivity and low device specification, and its two-phase content delivery strategy supports both the immediate Limpopo field study and the broader pilot deployment application that the AMEN team can use to continue research.

The evaluation methodology is grounded in the same literature that informs the design. Focus group sessions in Limpopo will combine an ISO 9241-11 usability assessment with a pre- and post-interaction nutritional knowledge assessment, supported by a co-design activity that gives participants meaningful agency in shaping the iterated design. Together these methods enable the project's research question on whether retained nutritional knowledge correlates with perceived usability, while also producing the qualitative insight needed to redesign the application for a more representative final build.

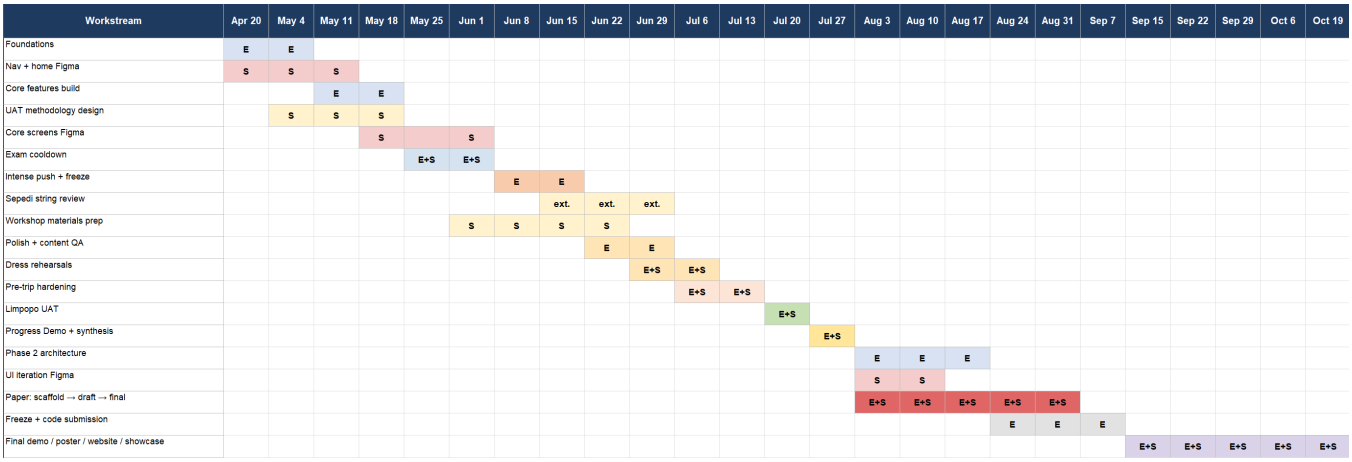


Figure 1: Gantt chart of the project timeline, 20 April – 26 October 2026. Work phases are colour-coded and key milestones marked.

We hope that this final build will be a valuable contribution to the research team and be a small but meaningful part of the journey to improving maternal and child well-being in rural Limpopo.

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